

Data Completion Model-based 2-D DOA Estimation for L-shaped array

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ABSTRACT

Based on the geometric structure of an L-shaped array, a self-supervised learning (SSL) framework based on transfer learning is proposed to achieve high-precision far-field two-dimensional (2-D) DOA estimation. The framework first pretrains a U-Net model using self-supervised learning to predict the missing regions with masked data. Subsequently, a transfer learning approach is employed to apply the representation ability of the pre-trained data completion model to a downstream 2-D DOA estimation problem. Then, the 2-D DOA estimation model is fine-tuned with a small amount of labeled data, which facilitates gridless DOA estimation. Experimental results show the robustness and effectiveness of the proposed method.

Index Terms— Self-supervised learning (SSL), data completion model, parameter estimation, direction of arrival.

1. INTRODUCTION

Direction of arrival (DOA) estimation, serving as a fundamental sensing mechanism, is very important to reconstruct spatial coordinates of the targets [1, 2], which has found numerous applications in radar and wireless communications [3–7]. For one-dimensional (1-D) DOA estimation, many high-resolution DOA estimation methods have been proposed, such as MUSIC [8] and ESPRIT [9]. Compared with 1-D DOA, two-dimensional (2-D) DOA estimation requires joint estimation of elevation and azimuth angles of incident signals, which can more accurately represent the spatial positioning characteristics [10, 11].

The performance of above traditional 2-D DOA estimation methods [12–14] typically relies on the ideal array data model. However, in practice, the ideal assumption is difficult to be established, especially in complex and changing environments, where the traditional methods may suffer from significant performance degradation.

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In recent years, with the development of deep learning (DL) [15–18], data-driven methods have gradually been introduced into the DOA estimation area. In general, most of the above deep neural network based techniques rely on supervised learning, which requires sufficient labeled training samples; however, obtaining large amounts of well-labeled DOA data for supervised learning is often time-consuming and difficult, especially in practical applications where the acquisition of labeled data is further complicated by the variability and complexity of the transmission environment [19–21]. Self-supervised learning (SSL) is an emerging learning approach, which aims to extract useful information from unlabeled data by designing auxiliary tasks to learn the internal information of the data, thus enabling the model to learn efficiently without human labelling. This SSL approach not only reduces the reliance on large amounts of labeled data, but also improves the generalization ability of the model, allowing it to better adapt to different environments. In addition, the performance of DOA estimation is further improved by incorporating migration learning to fine-tune downstream tasks.

In this paper, We propose a self-supervised U-Net framework pre-trained for data completion to learn robust spatial features. This pre-trained model is transferred via fine-tuning to a 2-D DOA estimation task, effectively leveraging limited labeled data. We then formulate DOA estimation as an end-to-end regression network that directly outputs continuous angles, avoiding discretization errors. Experiments show our method achieves superior accuracy and robustness in low-SNR and complex environments compared to supervised baselines.

2. SIGNAL MODEL

As shown in Fig. 1, consider an L-shaped array with inter-element spacing $d = \lambda/2$, where $M_x = M + 1$ and $M_y = N + 1$ sensors are placed on the x - and y -axes, respectively. For K uncorrelated far-field sources, the received data are

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{A}_x \mathbf{s}(t) + \mathbf{n}_x(t), \\ \mathbf{y}(t) &= \mathbf{A}_y \mathbf{s}(t) + \mathbf{n}_y(t), \end{aligned} \quad (1)$$

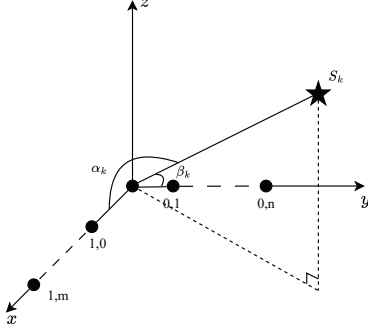


Fig. 1. Geometric structure of the L-shaped array.

where $\mathbf{A}_x = [\mathbf{a}_x(\alpha_1), \mathbf{a}_x(\alpha_2), \dots, \mathbf{a}_x(\alpha_K)]$ and $\mathbf{A}_y = [\mathbf{a}_y(\beta_1), \mathbf{a}_y(\beta_2), \dots, \mathbf{a}_y(\beta_K)]$. The steering vectors are

$$\begin{aligned} \mathbf{a}_x(\alpha_k) &= \left[1, e^{-j \frac{2\pi d}{\lambda} \cos \alpha_k}, \dots, e^{-j \frac{2\pi d}{\lambda} (M_x - 1) \cos \alpha_k} \right]^T, \\ \mathbf{a}_y(\beta_k) &= \left[1, e^{-j \frac{2\pi d}{\lambda} \cos \beta_k}, \dots, e^{-j \frac{2\pi d}{\lambda} (M_y - 1) \cos \beta_k} \right]^T. \end{aligned} \quad (2)$$

Here, α_k and β_k are defined as the angles of incidence relative to the x-axis and y-axis, respectively. The (α_k, β_k) and (θ_k, ψ_k) can be converted by

$$\cos \alpha_k = \sin \psi_k \cos \theta_k, \quad \cos \beta_k = \sin \psi_k \sin \theta_k. \quad (3)$$

where θ_k and ψ_k denote the azimuth and elevation angles under the spherical coordinate system with ψ_k measured from the positive z-axis. Accordingly, one may recover (θ_k, ψ_k) from the estimated (α_k, β_k) .

To suppress additive noise, we form the cross-correlation matrix

$$\mathbf{R}_{xy} \triangleq \mathbb{E}\{\mathbf{x}(t)\mathbf{y}^H(t)\} = \mathbf{A}_x \mathbf{R}_s \mathbf{A}_y^H, \quad (4)$$

where $\mathbf{R}_s = \mathbb{E}\{\mathbf{s}(t)\mathbf{s}^H(t)\}$ is diagonal for uncorrelated sources and the noise terms are assumed independent across the two arms such that $\mathbb{E}\{\mathbf{n}_x(t)\mathbf{n}_y^H(t)\} = \mathbf{0}$.

3. PROPOSED METHOD

This work aims to reduce the reliance on labeled data for 2-D DOA estimation with an L-shaped array. As illustrated in Fig. 2, the proposed framework consists of two stages.

Stage 1 (self-supervised pretext): we construct the cross-correlation matrix $\mathbf{R}_{xy}$ and form a two-channel real-valued representation. A masking operation is applied to simulate sensor failures by zeroing the corresponding rows/columns, and a completion network is trained to reconstruct the original $\mathbf{R}_{xy}$. The original unmasked matrix serves as the pseudo-label, enabling large-scale pre-training without manual annotations.

Stage 2 (transfer to DOA regression): the encoder and bottleneck of the pre-trained completion network is transferred to the target DOA estimation task. A lightweight regression

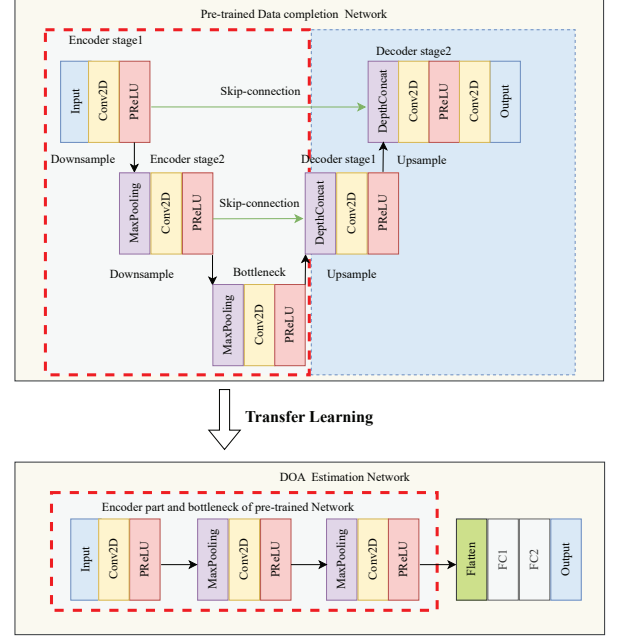


Fig. 2. Transfer learning from the pretext task to the DOA estimation target task.

head is appended and the network is fine-tuned using a small labeled set to output the estimated DOAs $\{(\alpha_k, \beta_k)\}_{k=1}^K$.

3.1. Data Representation and Pretext Task

The cross-correlation matrix $\mathbf{R}_{xy} \in \mathbb{C}^{M_x \times M_y}$ is first constructed and converted to a real-valued two-channel tensor:

$$\mathbf{X} = \text{cat}(\Re(\mathbf{R}_{xy}), \Im(\mathbf{R}_{xy})) \in \mathbb{R}^{M_x \times M_y \times 2}. \quad (5)$$

where $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts, respectively, and $\text{cat}(\cdot)$ concatenates along the channel dimension. For numerical stability, the input tensor is normalized by its Frobenius norm:

$$\tilde{\mathbf{X}} = \mathbf{X} / \|\mathbf{X}\|_F. \quad (6)$$

To simulate sensor failures on the two arms of the L-shaped array, we define two index sets, Ω_x and Ω_y . Here, $\Omega_x \subseteq \{1, 2, \dots, M_x\}$ and $\Omega_y \subseteq \{1, 2, \dots, M_y\}$ represent the masked sensor indices on the x-axis and y-axis, respectively. Masking sensor failures can be equivalently implemented as row and column masking on $\tilde{\mathbf{X}}$. Define two diagonal masking matrices $\mathbf{P}_x \in \mathbb{R}^{M_x \times M_x}$ and $\mathbf{P}_y \in \mathbb{R}^{M_y \times M_y}$ as

$$(\mathbf{P}_x)_{m,m} = \begin{cases} 0, & m \in \Omega_x, \\ 1, & m \notin \Omega_x, \end{cases} \quad (\mathbf{P}_y)_{n,n} = \begin{cases} 0, & n \in \Omega_y, \\ 1, & n \notin \Omega_y. \end{cases} \quad (7)$$

where $m \in \{1, \dots, M_x\}$ and $n \in \{1, \dots, M_y\}$. Then, the masking operator $\mathcal{M}(\cdot; \Omega_x, \Omega_y)$ is defined by left/right multiplication:

$$\bar{\mathbf{X}} = \mathcal{M}(\tilde{\mathbf{X}}; \Omega_x, \Omega_y) = \left(\mathbf{P}_x \tilde{\mathbf{X}}^{(1)} \mathbf{P}_y, \mathbf{P}_x \tilde{\mathbf{X}}^{(2)} \mathbf{P}_y \right), \quad (8)$$

where $\tilde{\mathbf{X}}^{(1)}$ and $\tilde{\mathbf{X}}^{(2)}$ denote the first and second channels of $\tilde{\mathbf{X}}$, and the same $\mathbf{P}_x, \mathbf{P}_y$ are applied to both channels. As a result, all entries in rows indexed by Ω_x and columns indexed by Ω_y are set to zero in the correlation representation, consistent with sensor failures on the corresponding array arms.

In the implementation, Ω_x and Ω_y are sampled uniformly at random with fixed mask ratios and are resampled across training samples to cover diverse failure patterns. The same masking strategy is adopted in both self-supervised pre-training and downstream fine-tuning.

The pretext task is to reconstruct the original $\tilde{\mathbf{X}}$ from $\bar{\mathbf{X}}$. Therefore, the unmasked $\tilde{\mathbf{X}}$ serves as the pseudo-label, enabling self-supervised pre-training without manual annotations.

In the target task, the model outputs the 2-D DOAs $\mathbf{z} = \{(\alpha_k, \beta_k)\}_{k=1}^K$. The completion network is first pre-trained using a large unlabeled set, and then fine-tuned on a much smaller labeled set for DOA regression.

3.2. Pretext Network for Cross-correlation Completion

A U-Net-like encoder-decoder network is employed to address the masked completion pretext task. Given the masked input $\bar{\mathbf{X}}$ in (8), the completion network outputs a reconstructed tensor

$$\hat{\mathbf{X}} = \mathcal{F}_{\text{comp}}(\bar{\mathbf{X}}), \quad (9)$$

which is trained to approximate the pseudo-label $\tilde{\mathbf{X}}$ in (6). Both $\tilde{\mathbf{X}}$ and $\hat{\mathbf{X}}$ are real-valued tensors in $\mathbb{R}^{M_x \times M_y \times 2}$, corresponding to the stacked real and imaginary parts of the cross-correlation matrix.

The encoder progressively extracts multi-scale correlation patterns via down-sampling, while the decoder restores the resolution through up-sampling. Skip connections are introduced to fuse low-level local structures with high-level global features, which helps recover occluded entries and alleviates over-smoothing.

Let $\mathbf{W} \in \{0, 1\}^{M_x \times M_y}$ denote the entry-wise mask induced by (Ω_x, Ω_y) , where $W_{m,n} = 1$ if $m \in \Omega_x$ or $n \in \Omega_y$, and $W_{m,n} = 0$ otherwise. The same mask is applied to both channels of the two-channel input. The pretext network is trained by minimizing the masked reconstruction loss:

$$\mathcal{L}_{\text{pre}} = \left\| \mathbf{W} \odot (\hat{\mathbf{X}} - \tilde{\mathbf{X}}) \right\|_F^2, \quad (10)$$

where $\odot$ denotes the Hadamard (element-wise) product and $\|\cdot\|_F$ denotes the Frobenius norm.

By learning to restore the occluded entries of $\mathbf{R}_{xy}$ under diverse masking patterns, the encoder captures structural patterns of array correlation, which provides transferable representations for the downstream DOA regression task.

3.3. Transfer to 2-D DOA Regression

After pre-training, the encoder and bottleneck learned from the completion task are transferred to the 2-D DOA estimation task, as shown in Fig. 2. A lightweight regression head $\mathcal{H}(\cdot)$ is appended to predict the DOA parameters:

$$\hat{\mathbf{z}} = \mathcal{H}\left(\text{Enc}(\hat{\mathbf{X}})\right), \quad \mathbf{z} = \{(\alpha_k, \beta_k)\}_{k=1}^K, \quad (11)$$

where $\text{Enc}(\cdot)$ denotes the transferred encoder and $\hat{\mathbf{z}}$ is the estimated DOA set.

To avoid permutation ambiguity in multi-source regression, we adopt an ordered-output formulation during training. For each sample, the ground-truth DOA pairs are deterministically sorted according to the α angle in ascending order, and the labels are arranged accordingly (e.g., for $K = 2$, (α_1, β_1) and (α_2, β_2) satisfy $\alpha_1 \leq \alpha_2$). The regression head is trained to output DOAs in the same order, leading to a one-to-one correspondence between each output and its assigned label; therefore, the regression loss is unambiguous and does not require additional set matching.

A two-step fine-tuning strategy is adopted: (i) the transferred encoder is frozen and only the regression head is trained on a small labeled set; (ii) the encoder is then unfrozen and the entire network is fine-tuned for improved accuracy. The regression loss is defined as

$$\mathcal{L}_{\text{doa}} = \|\hat{\mathbf{z}} - \mathbf{z}\|_1, \quad (12)$$

where $\|\cdot\|_1$ denotes the ℓ_1 -norm. This loss promotes robust angle regression under limited supervision.

4. EXPERIMENTS

4.1. Experimental Setup

An L-shaped array with $M = N = 7$ is considered, where there are 8 sensors on each of the x - and y -axes. Unless otherwise specified, the number of uncorrelated sources is set to $K = 2$. The candidate angle search ranges are $\alpha \in (20^\circ, 120^\circ)$ and $\beta \in (60^\circ, 160^\circ)$. To reduce candidate pairs while keeping full coverage, the inter-source angular separations in both angles are restricted. Specifically, the separation for α is selected from $\Delta\alpha \in \{2^\circ, 4^\circ, \dots, 10^\circ\}$, and the separation for β is selected from $\Delta\beta \in \{10^\circ, 20^\circ\}$. For each chosen $\Delta\alpha$, $\alpha_1 \in \{20^\circ, 22^\circ, \dots, 120^\circ - \Delta\alpha\}$ is sampled and $\alpha_2 = \alpha_1 + \Delta\alpha$; similarly, for each chosen $\Delta\beta$, $\beta_1 \in \{60^\circ, 62^\circ, \dots, 160^\circ - \Delta\beta\}$ is sampled and $\beta_2 = \beta_1 + \Delta\beta$.

To simulate sensor failures, masked samples are generated using the random masking strategy defined in Sec. 3.1; the

mask ratio is fixed and results are averaged over multiple random masking realizations.

For the self-supervised pretext task, the training set $\mathcal{J}_1$ is generated with SNR ranging from -5 dB to 20 dB with a 5 dB step. For each SNR, $80,000$ samples are generated with the snapshot number $T = 512$, thus $\mathcal{J}_1$ contains $D_1 = 6 \times 80,000 = 480,000$ training samples in total. For the downstream DOA regression task, the fine-tuning set $\mathcal{J}_2$ contains $20,000$ labeled samples per SNR, thus $D_2 = 6 \times 20,000 = 120,000$.

Both pretext and target-task datasets are randomly split into 80% for training and 20% for validation. The Adam optimizer is used with batch size 64 and an initial learning rate of 10^{-3} . If the validation loss does not improve within 5 consecutive epochs, the learning rate is multiplied by 0.7 . For transfer learning, the encoder and bottleneck are first frozen and only the DOA regression head is trained on $\mathcal{J}_2$; then the whole network is unfrozen for end-to-end fine-tuning.

The proposed SSL transfer method is compared with supervised learning (SL) baselines trained under different data availability: (i) $\text{SL}+\mathcal{J}_2+\text{mask}$, (ii) $\text{SL}+\mathcal{J}_1+\text{mask}$, (iii) $\text{SL}+\mathcal{J}_2+\text{nomask}$, and (iv) $\text{SL}+\mathcal{J}_1+\text{nomask}$.

4.2. DOA Estimation Performance

The 2-D DOA estimation accuracy is evaluated using RMSE over Monte Carlo trials. Let $\hat{\alpha}_k^{(h)}$ and $\hat{\beta}_k^{(h)}$ denote the estimates of (α_k, β_k) in the h -th Monte Carlo trial for the k -th source. The 2-D RMSE is computed as

$$\text{RMSE} = \sqrt{\frac{1}{NK} \sum_{h=1}^N \sum_{k=1}^K \left(\left(\hat{\alpha}_k^{(h)} - \alpha_k \right)^2 + \left(\hat{\beta}_k^{(h)} - \beta_k \right)^2 \right)}. \quad (13)$$

RMSE is first evaluated under different SNRs with $T = 512$ snapshots and 500 Monte Carlo trials. Fig. 3a reports an *on-grid* case where two signals are incident from $(30^\circ, 86^\circ)$ and $(33^\circ, 96^\circ)$, respectively. As expected, RMSE decreases as SNR increases. Due to limited labeled data and missing sensors, $\text{SL}+\mathcal{J}_2+\text{mask}$ yields the poorest performance. In contrast, the proposed SSL transfer method substantially improves the estimation accuracy and achieves results comparable to $\text{SL}+\mathcal{J}_1+\text{nomask}$, demonstrating that effective representations can be learned from large-scale unlabeled data.

An *off-grid* case is further considered with two signals from $(86^\circ, 106.1^\circ)$ and $(90.5^\circ, 116^\circ)$. Fig. 3b shows the RMSE performance with SNR ranging from -5 dB to 25 dB. The proposed method consistently outperforms supervised baselines trained on small labeled sets and exhibits superior generalization in off-grid estimation.

Next, RMSE is evaluated versus the number of snapshots. For the *on-grid* case in Fig. 4a, SNR is fixed to 0 dB and the number of snapshots is set to $T \in \{250, 500, 1000, 1500, 2000, 2500\}$. The proposed SSL transfer method achieves stable performance with fewer snapshots and remains close to

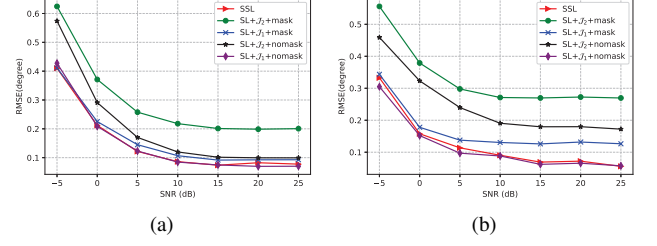


Fig. 3. RMSE versus SNR: (a) two signals from $(30^\circ, 86^\circ)$ and $(33^\circ, 96^\circ)$, (b) two signals from $(86^\circ, 106.1^\circ)$ and $(90.5^\circ, 116^\circ)$.

the strong baseline $\text{SL}+\mathcal{J}_1+\text{nomask}$, while methods trained with smaller labeled sets and masked data degrade noticeably.

For the off-grid case in Fig. 4b, SNR is fixed to 5 dB and T is varied in the same range. The proposed method maintains clear advantages under the off-grid setting, indicating better generalization under limited snapshots and missing sensors.

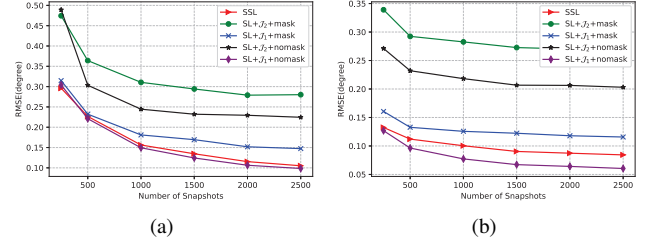


Fig. 4. RMSE versus the number of snapshots: (a) two signals from $(30^\circ, 86^\circ)$ and $(33^\circ, 96^\circ)$, (b) two signals from $(86^\circ, 106.1^\circ)$ and $(90.5^\circ, 116^\circ)$.

5. CONCLUSIONS

This paper proposes a self-supervised transfer framework for 2-D DOA estimation with limited labeled data. A masking-based completion pretext task is introduced to learn transferable representations from large-scale unlabeled samples. The pre-trained encoder is subsequently fine-tuned for DOA regression, enabling accurate estimation under missing-sensor settings. Experiments under varying SNRs and snapshot numbers demonstrate that the proposed method consistently outperforms supervised baselines trained on small labeled sets while achieving competitive performance. Moreover, improved generalization is observed in off-grid scenarios.

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